Phase-1 Report: Accelerating Goal-Conditioned Reinforcement Learning via Contrastive Learning in JAX

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Repo: github.com/JB243/ECE567_W2026_Team5_GCRL

1. Environments

Goal-conditioned reinforcement learning (GCRL) addresses the issue of sparse reward by conditioning the policy on a target goal state. We evaluate GCRL methods on four environments from the JaxGCRL benchmark: **Reacher**, **Humanoid**, **Ant U-Maze**, and **Ant Soccer**. These four environments were selected because they span both locomotion and manipulation tasks with varying control complexity and exploration difficulty.

Reacher. A 2D robotic manipulation task in which a two-segment arm must move its end effector to a target position sampled uniformly from a disk around the initial state.

Humanoid. A high-dimensional locomotion task where a humanoid agent must move toward a goal while maintaining balance and coordinating full-body motion.

Ant U-Maze. A navigation task built on the quadruped Ant robot, where the agent must reach a goal while navigating around walls in a U-shaped maze.

Ant Soccer. An object-interaction task in which the Ant agent must push a ball-like object into a goal region. The target is sampled around the environment, and the position of the movable sphere is randomized along the line between the agent and the goal.

2. Baselines

To evaluate performance across the selected environments, we reproduced six baseline methods: **CRL**, **SAC**, **SAC+HER**, **TD3**, **TD3+HER**, and **PPO**. These baselines represent a mixture of standard continuous-control RL algorithms, goal-conditioned extensions using hindsight relabeling, and the contrastive goal-conditioned approach.

PPO (Proximal Policy Optimization). An on-policy policy-gradient method widely used in continuous control. It optimizes a clipped surrogate objective to stabilize updates.

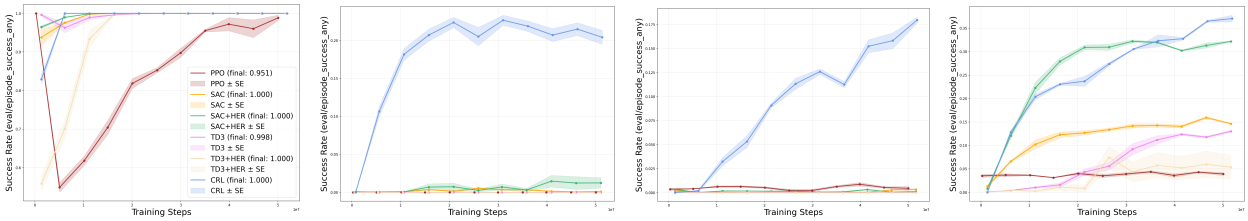
SAC (Soft Actor-Critic). An off-policy actor-critic algorithm for continuous control that optimizes a stochastic policy with an entropy bonus to encourage exploration and avoid premature convergence.

SAC+HER. SAC combined with Hindsight Experience Replay (HER), which relabels failed episodes using goals that were actually achieved later in the same trajectory to provide denser signal.

TD3 (Twin Delayed Deep Deterministic Policy Gradient). An off-policy actor-critic algorithm for continuous control using twin Q-networks, delayed policy updates, and target policy smoothing.

TD3+HER. TD3 combined with HER for improved learning in sparse-reward goal-reaching tasks.

CRL (Contrastive Reinforcement Learning). The main method of interest in this benchmark. CRL learns goal-reaching behavior by treating future states from the agent's trajectories as contrastive learning targets, and so it learns a representation of reachability instead of relying only on sparse reward signals.



(a) Reacher (b) Humanoid (c) Ant Ball (d) Ant U-Maze

Figure 1: Success rate comparison across four environments.

3. Results

We evaluate performance using **success rate**, defined as the fraction of evaluation episodes in which the agent reaches the goal within a specified threshold. All results are aggregated as the interquartile mean over 10 random seeds. As shown in Figures 1a, 1b, 1c, and 1d, CRL consistently achieves the best performance across all four environments, while PPO performs the worst.

The results can be categorized into three regimes based on task difficulty and reward sparsity. In **Reacher**, all methods perform similarly well because the task is short-horizon easy to explore (Fig. 1a). In **Humanoid** and **Ant Ball**, only CRL achieves meaningful performance, which suggests that its use of intermediate goals significantly improves exploration in long-horizon tasks (Figs. 1b and 1c). In **Ant U-Maze**, multiple methods achieve non-trivial performance. While HER improves learning through goal relabeling, CRL further outperforms other approaches by learning a dense notion of reachability from trajectory data (Fig. 1d).

4. Reproducibility

Our reproduction supports the main claims of the JaxGCRL benchmark. In particular, JaxGCRL achieves approximately 10M environment steps in under 10 minutes on a single GPU, confirming its strong wall-clock efficiency advantage relative to conventional PyTorch-based pipelines. In terms of task performance, CRL consistently outperforms the standard baselines on the harder sparse-reward environments. This supports the claim that future-state hindsight goals provide a powerful inductive bias for sparse-reward exploration.

However, we observed discrepancies relative to the paper. First, in **Reacher**, PPO occasionally achieving near-perfect performance after extended training, whereas the paper reports substantially weaker PPO performance. On the more complex environments, however, PPO consistently fails to learn useful policy, which aligns with the conclusion of the paper that PPO is poorly suited to these sparse-reward tasks.

Second, we identified inconsistencies between the configurations reported in the paper and those used in the released implementation for CRL on **Ant U-Maze**. According to Table 2, the paper specifies `energy_fn = "l2"`, `contrastive_loss_fn = "sym_infonce"`, and a policy learning rate of 6×10^{-4}. However, using these settings in our reproduction caused TD3+HER to eventually outperform CRL, which contradicts the reported benchmark results. To obtain results more consistent with the paper, we instead adopted the default configuration in the released code, namely `energy_fn = "norm"`, `contrastive_loss_fn = "fwd_infonce"`, and a policy learning rate of 3×10^{-4}. These findings suggest that CRL's performance is highly sensitive to implementation details that are not fully specified in the paper.

References

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